PR-1 Elka 1

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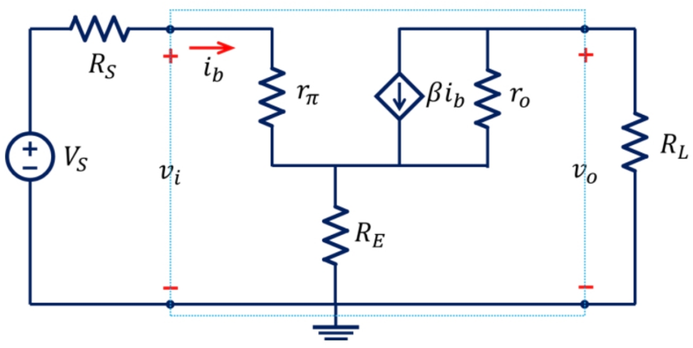
Soal 1 Model Penguat

Pada rangkaian berikut ini akan dicari $\frac{v_o}{v_i}$ dan $\frac{v_o}{v_s}$.

a) Gunakan cara yang diajarkan pada 2 port network untuk memodelkan penguat yang ditandai dengan kotak garis biru putus-putus sebagai salah satu (saja) dari penguat tegangan, penguat arus, penguat transkonduktansi, atau penguat transresistansi.

b) Gantikan rangkaian dengan model pada (a) dan carilah $\frac{v_o}{v_i}$ dan $\frac{v_o}{v_s}$



a)

Mesh 1

$$V_i = i_b r_\pi + R_E(i_b - i_o)$$
$$V_i = i_b(r_\pi + R_E) - i_o R_E$$

Mesh 2

$$V_o + i_o r_o + r_o \beta i_b + R_E(i_o - i_b) = 0$$
$$V_o = -i_o(r_o + R_E) + i_b(R_E - \beta r_o)$$

$i_o = 0$

$$V_i = i_b(r_\pi + R_E) = i_b R_i \qquad R_i = r_\pi + R_E$$

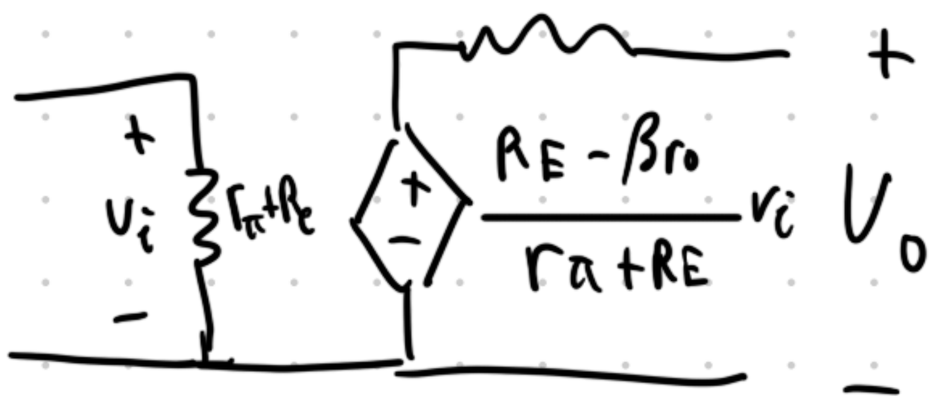
$$V_o = i_b(R_E - \beta r_o)$$
$$V_o = \frac{V_i}{(r_\pi + R_E)}(R_E - \beta r_o)$$
$$V_o = V_i\left(\frac{R_E - \beta r_o}{r_\pi + R_E}\right) = V_i A_{vo}$$
$$A_{vo} = \frac{R_E - \beta r_o}{r_\pi + R_E}$$

$V_i = 0 \quad i_o = -i_x \quad V_o = V_x$

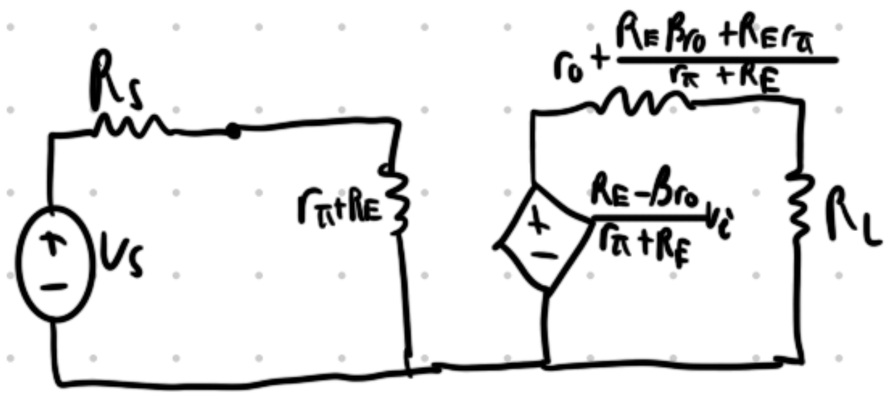
$$0 = i_b(r_\pi + R_E) - i_o R_E$$
$$i_o R_E = i_b(r_\pi + R_E)$$
$$i_b = i_o\left(\frac{R_E}{r_\pi + R_E}\right)$$

$$V_x = -i_o(r_o + R_E) + i_o\left(\frac{R_E}{r_\pi + R_E}\right)(R_E - \beta r_o)$$
$$V_x = i_o\left(\frac{R_E(R_E - \beta r_o)}{r_\pi + R_E} - r_o - R_E\right)$$
$$V_x = i_o\left(\frac{R_E(R_E - \beta r_o)}{r_\pi + R_E} - \frac{R_E(r_\pi + R_E)}{r_\pi + R_E} - r_o\right)$$
$$V_x = i_o\left(\frac{\cancel{R_E^2} - R_E\beta r_o - R_E r_\pi - \cancel{R_E^2}}{r_\pi + R_E} - r_o\right)$$
$$V_x = -i_x\left(\frac{-R_E\beta r_o - R_E r_\pi}{r_\pi + R_E} - r_o\right)$$
$$\frac{V_x}{i_x} = R_o = r_o + \frac{R_E\beta r_o + R_E r_\pi}{r_\pi + R_E}$$

$$r_o + \frac{R_E\beta r_o + R_E r_\pi}{r_\pi + R_E}$$



b)

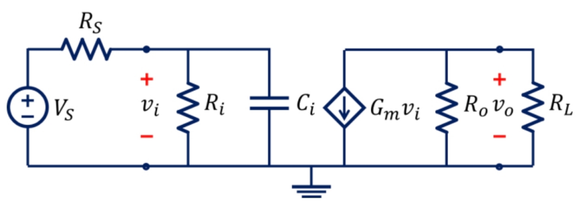


$$\frac{V_o}{V_i} = A_{vo}\left(\frac{R_L}{r_o + \frac{R_E\beta r_o + R_E r_\pi}{r_\pi + R_E} + R_L}\right) = \left(\frac{R_E - \beta r_o}{r_\pi + R_E}\right)\left(\frac{R_L}{r_o + \frac{R_E\beta r_o + R_E r_\pi}{r_\pi + R_E} + R_L}\right)$$

$$\frac{V_o}{V_s} = A_{vo}\left(\frac{r_\pi + R_E}{R_s + r_\pi + R_E}\right)\left(\frac{R_L}{r_o + \frac{R_E\beta r_o + R_E r_\pi}{r_\pi + R_E} + R_L}\right) = \left(\frac{R_E - \beta r_o}{R_s + r_\pi}\right)\left(\frac{r_\pi + R_E}{R_s + r_\pi + R_E}\right)\left(\frac{R_L}{r_o + \frac{R_E\beta r_o + R_E r_\pi}{r_\pi + R_E} + R_L}\right)$$

Soal 2 Respons Frekuensi Penguat (Orde Satu)

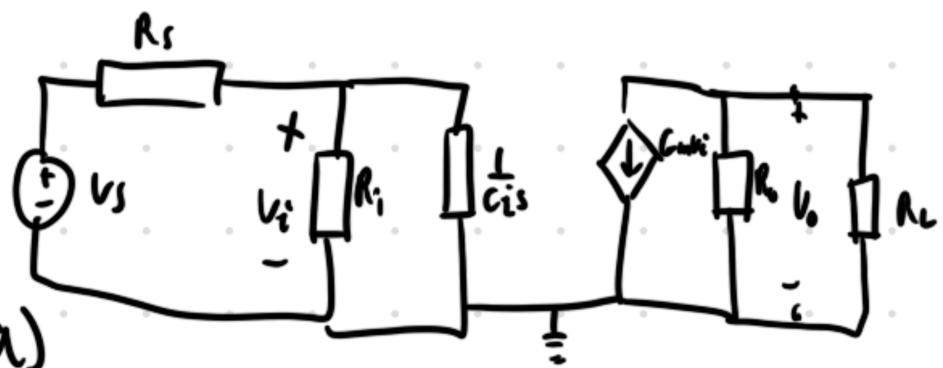
Berikut ini adalah model sebuah penguat dengan respons HPF.



a) Tentukan frekuensi cut-off rangkaian tersebut dinyatakan dalam variabel nilai komponennya.

b) Bila rangkaian memiliki $R_S = 300\Omega$, $R_i = 2k\Omega$, $R_o = 10k\Omega$, $R_L = 1k\Omega$, $C_i = 10pF$, dan $G_m = 50\text{mA/V}$. Tentukan penguatan pada frekuensi rendah (passband) dan frekuensi cut-offnya.

c) Gunakan SPICE untuk mengkonfirmasi jawaban.



a)

$$\frac{V_i}{V_s} = \frac{\left(\frac{1}{R_i} + C_i s\right)^{-1}}{R_s + \left(\frac{1}{R_i} + C_i s\right)^{-1}} = \frac{\frac{R_i}{1 + C_i s R_i}}{R_s + \left(\frac{R_i}{1 + C_i s R_i}\right)} = \frac{R_i}{R_s(1 + C_i s R_i) + R_i}$$

$$= \frac{R_i}{R_s + R_i + C_i s R_i R_s}$$

$$= \frac{\frac{R_i}{R_s + R_i}}{1 + \left(\frac{R_i R_s}{R_s + R_i}\right) C_i s}$$

$$\frac{V_o}{V_i} = -G_m\left(\frac{1}{R_o}+\frac{1}{R_L}\right)^{-1} = -G_m\frac{R_oR_L}{R_o+R_L}$$

$$\frac{V_o}{V_s} = \frac{V_o}{V_i}\cdot\frac{V_i}{V_s} = -G_m\frac{R_oR_L}{R_o+R_L}\cdot\frac{\frac{R_i}{R_s+R_i}}{1+\frac{R_iR_s}{R_s+R_i}C_iS}$$

$$H(S) = \frac{-G_m\frac{R_oR_LR_i}{(R_o+R_L)(R_s+R_i)}}{1+S\left(\frac{R_iR_s}{R_s+R_i}\right)C_i} = \frac{A_{vo}}{1+\frac{S}{\omega_c}}$$

$$\frac{1}{\omega_c} = \frac{R_iR_s}{R_s+R_i}C_i$$

$$\omega_c = \frac{R_s+R_i}{R_iR_sC_i} \qquad f_c = \frac{R_s+R_i}{2\pi R_iR_sC_i}$$

b) $A_{vo} = -G_m\frac{R_oR_LR_i}{(R_o+R_L)(R_s+R_i)}$

$$= -(50\cdot10^{-3})\frac{10\cdot10^3\cdot10^3\cdot2\cdot10^3}{(10\cdot10^3+10^3)(300+2\cdot10^3)}$$

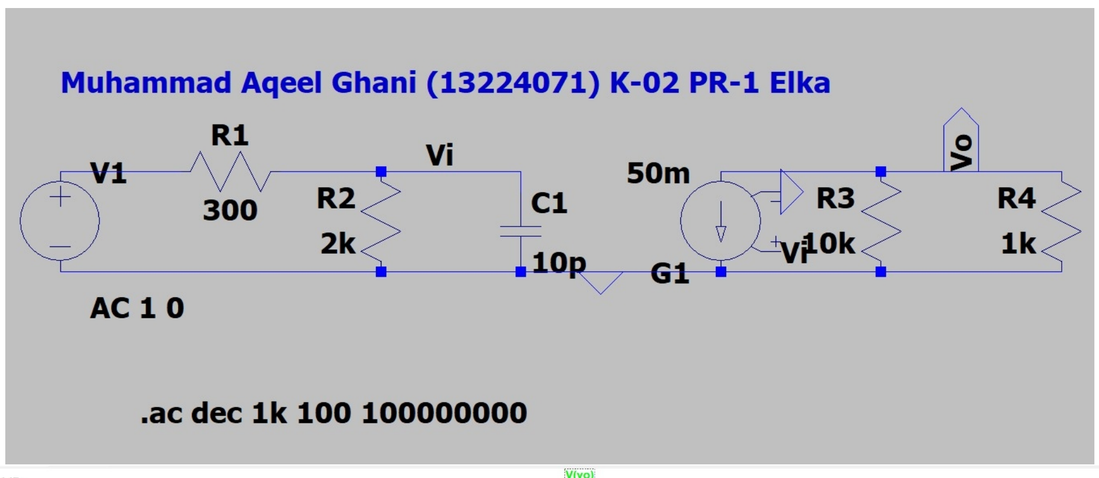
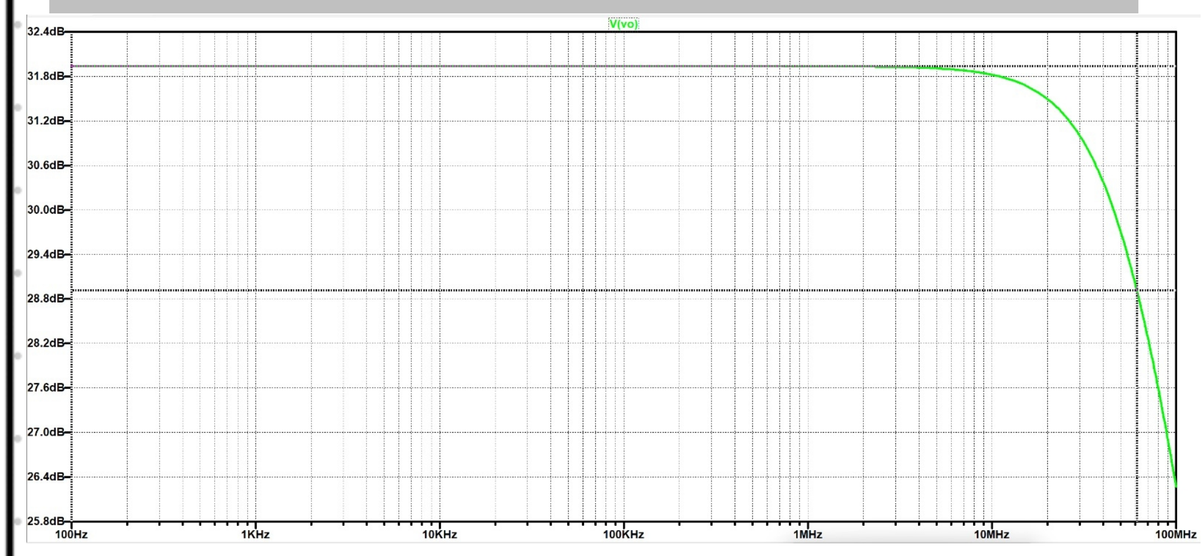
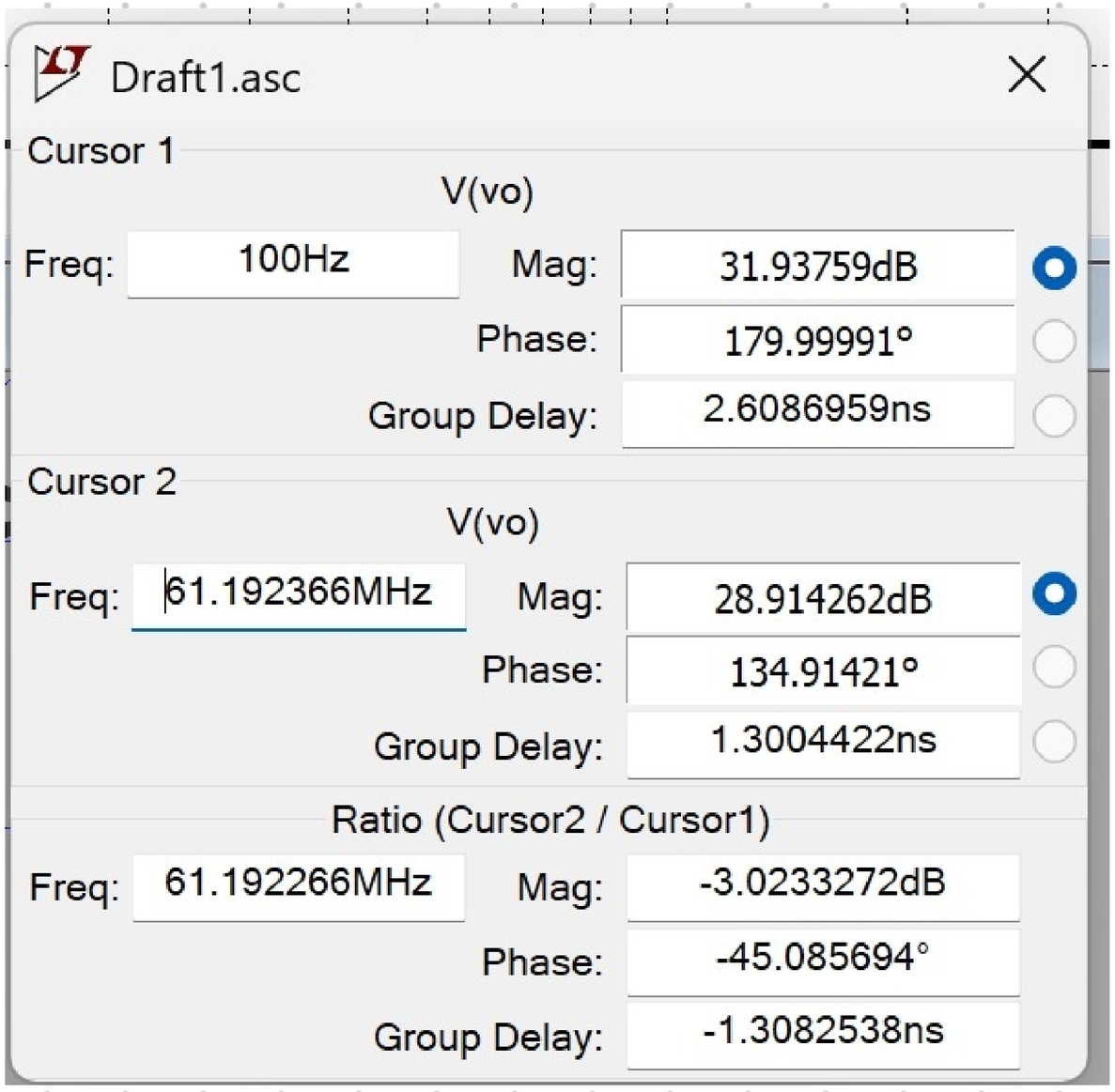
$$= -\frac{10000}{253} = -39,5256917$$

$$A_{db} = 20\log\left(\frac{10000}{253}\right) = 31,93758958\ dB$$

$$f_c = \frac{R_s+R_i}{2\pi R_iR_sC_i} = \frac{300+2\cdot10^3}{2\pi\cdot2\cdot10^3\cdot300\cdot10\cdot10^{-12}}$$

$$= 61009394,85\ Hz$$
$$= 61,01\ MHz$$

Penguatan saat frekuensi cut-off

$$A_{fc} = 20\log\left(\frac{10000}{253}\div\sqrt{2}\right) = 28,92728962\ dB$$

c)